

Indian Statistical Institute

BSDS, First Year, First Semester Examination, 2025-26

Introduction to Computing

Answer Keys

1. (a) 10
(b) [10]
(c) ten
(d) ['t', 'e', 'n']
(e) 10.0

2. Here follows the object-oriented version.

```
class Queue():
    def __init__(self):
        self.QueueDS = []
        self.SizeQueueDS = 10
        print ("Queue initialized")
    def QueueINSERT(QueueDS, Data):
        if len(QueueDS) < SizeQueueDS:
            QueueDS.insert(0, Data)
        else:
            print ("Queue overflow!!!")
    def QueueDELETE(QueueDS):
        if len(QueueDS) > 0:
            Data = QueueDS.pop()
            return Data
        else:
            print ("Queue underflow!!!")
    def QueueDISPLAY(QueueDS):
        print ("The elements in the queue are: ")
        for Element in QueueDS:
            print(Element)
    def __del__(self):
        print ("Queue destructed")

Queue q
q.initialize()
del q
```

3. (a) Here follows a correct version of the program.

```
def Bsearch(array, target):
    low = 0
    high = len(array) - 1
    while low <= high:
        mid = (low + high)//2
        if array[mid] == target:
            return mid
        elif array[mid] < target:
```

```

        low = mid + 1
    else:
        high = mid - 1
    return -1

```

The data format of the argument has to be a list.

(b) The purpose is to generate the power set of the characters in a string in the form of list of sublists traversed from the end of the string and ordered by the length of elements in the sublists.

4. Here follows an efficient version. Note that there can be many different (even better) such versions.

```

n = int(input())
is_prime = [True] * (n+1)
is_prime[0:2] = [False, False]
for i in range(2, int(n**0.5)+1):
    if is_prime[i]:
        for j in range(i*i, n+1, i):
            is_prime[j] = False
primes = [i for i in range(n+1) if is_prime[i]]
print ("Primes up to", n, ":", primes)

```

5. (a) 4

(b)

```

      11
     /
    5
   /
  7
 \
  3
 \
  2

```

This is unique and hence no alternate construction is possible.

6. (i) TRUE

(ii) TRUE

(iii) TRUE

(iv) FALSE

(v) TRUE

7. (a) Here follows the runtime complexities.

(i) $O(n^2)$

(ii) $\Theta(n^2)$

(b) Heapsort